CHAPTER 15

Architecture Maturity Models



This chapter is part of the GOLD reading tour. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

An Enterprise Architecture maturity model serves as a framework to assess how well an organization's architecture aligns with its business objectives. By outlining an ideal Enterprise Architecture implementation – typically conforming to a specific architecture framework – a maturity model framework provides a benchmark to evaluate the effectiveness of architectural processes. This assessment not only highlights the current maturity level but also offers a structured pathway for enhancement, guiding organizations toward improved alignment between their IT infrastructure and business goals.

In the "Architecture Maturity Models" Series Guide, the TOGAF Standard refers to two applicable methods for measuring architecture maturity. These are the Architecture Capability Maturity Model (ACMM) and Capability Maturity Model Integration (CMMI). Organizations can also develop their own organization-specific model to fit their unique circumstances.

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15.1. Determining Architecture Maturity

In the Preliminary Phase of the Architecture Development Method, a maturity model is often used to measure and provide insight into the maturity of the internal Architecture Capability. The "Architecture Maturity Models" Series Guide discusses two common frameworks for measuring maturity. One is the Architecture Capability Maturity Model framework. It has a structure in which the core components of an architecture process are represented and measured, often in relation to the organization's IT. The purpose of using such a framework is to identify weaknesses and provide a step-by-step approach to improving them.

The second is the Capability Maturity Model Integration framework. This framework places more emphasis on maturing the Architecture Capability in order to achieve organizational goals. This makes this type of maturity framework more appropriate for use in an Enterprise Architecture environment.

Enterprise Architecture is considered a strategic business management tool, not a technical instrument [9].

Enterprise Architecture is a strategic discipline that focuses on translating an organization's strategy into its execution and impact.

15.1.1. Architecture Capability Maturity Model

An Architecture Capability Maturity Model consists of three parts:

- The model itself
- The elements (characteristics) of the architecture processes, grouped into different maturity levels
- A scoring mechanism

The first two parts explain the maturity levels and the corresponding architecture elements for each level. The third section is designed to help determine the current level of maturity. This level can then be reported on.

The Architecture Capability Maturity Model (see Table 15-1) considers the following nine architecture *elements* when assessing architecture maturity.

Table 15-1. *Elements of ACMMs*

Element	Description
Architecture process	Refers to the extent to which architecture operations are systematically defined, meticulously documented, and consistently followed.
Architecture development	Encompasses the methods and tools used to create architecture solutions. Critical evaluation includes assessing how well they function.
Business linkage	Refers to the degree of alignment between IT investments and core business objectives.
Senior management involvement	How effectively leadership involvement is aiding the development and maintenance of architecture initiatives.
Business unit participation	The degree to which different business units and departments are involved in the architecture process.
Architecture communication	How effectively architecture processes are communicated throughout the organization.
Architecture governance	Involves the establishment of mechanisms and structures to oversee and guide architecture efforts.
IT security	Addresses the organization's ability to effectively meet cybersecurity requirements.
IT investment and acquisition strategy	Focuses on the judicious allocation of resources to existing systems and the acquisition of new capabilities.

Architecture Process: The architecture process refers to the extent to which architecture operations are systematically defined, meticulously documented, and consistently followed. In organizations where architectural practices are well integrated, these processes are characterized by rationality, repeatability, and clarity, thereby facilitating efficient execution and alignment with organizational goals.

Architecture Development: Architecture development encompasses the methods and tools used to create architecture solutions. Critical evaluation includes assessing the effectiveness of these methods and tools, determining the standards they are intended to meet, and verifying adherence to established best practices. Such scrutiny ensures that the architecture framework remains robust, adaptable, and aligned with industry benchmarks.

Business Linkage: Business linkage refers to the degree of alignment between IT investments and core business objectives. A well-structured architecture approach ensures that the IT infrastructure not only supports but also enhances business strategies, delivering measurable return on investment and driving business growth.

Senior Management Involvement: Senior management involvement is critical to the development and maintenance of architecture initiatives. Effective leadership provides the necessary sponsorship and resources to protect architecture projects from potential derailment due to resource constraints or lack of strategic direction.

Business Unit Participation: The involvement of various business units and departments in the architecture process indicates the depth of integration within the organization. A high level of participation indicates a collaborative environment where architecture practices are embraced throughout the organization, promoting coherence and optimal performance.

Architecture Communication: Effective communication of architecture processes within the organization ensures that all stakeholders have a clear understanding of their roles and overall goals. Transparent dissemination of information promotes alignment, facilitates informed decision-making, and increases the overall effectiveness of architecture efforts.

Architecture Governance: Architecture governance involves the establishment of mechanisms and structures to oversee and guide architecture efforts. Robust governance frameworks assign clear roles and responsibilities, ensure compliance and accountability, and foster innovation within architecture projects.

IT Security: IT security, in the context of architecture, addresses the organization's ability to effectively meet cybersecurity requirements. Mature architecture practices incorporate comprehensive security policies and cost-effective measures to safeguard sensitive data and protect IT assets from potential threats.

IT Investment and Acquisition Strategy: IT investment and acquisition strategy focuses on the judicious allocation of resources to existing systems and the acquisition of new capabilities. A strategic and economical approach to the use of IT resources ensures that investments are aligned with organizational goals and contribute to sustainable value creation.

In addition to the architecture process characteristics, an Architecture Capability Maturity Model has *six* maturity levels against which the Architecture Capability is measured (see Table 15-2). In practice, the lowest level (Level 0) is often omitted. The reason for this is that Level 0 indicates that nothing is being done with respect to architecture at all. In this case, there is no need to complete a maturity model.

Table 15-2. *Maturity Levels of ACCMs*

Level	Name	Description
0	None	Indicates a complete lack of architecture development.
1	Initial	Elements have been identified as worth developing but not yet have any processes defined.
2	Under development	Elements have begun being issued processes, tools, and methods for the purpose of architecture optimization.
3	Defined	Elements are in active development, having been given a standardized and content-complete set of architecture practices.
4	Managed	Elements have been developed to the extent that performance metrics have come into focus.
5	Measured	Element have been optimized as fully as the architecture team could envision, making it an example for other elements.

None: Indicates a complete lack of architecture development.

Initial: At this preliminary stage, the architecture element has been recognized as a candidate for further development, although no formalized processes have been established. Any changes or experimentation to date has been done in an unstructured and isolated manner, without broader organizational visibility or buy-in.

Under Development: At this stage, initial methodologies, tools, and frameworks are being implemented to support the development of the architectural element. Preliminary documentation may exist, but activities remain informal and efforts have not yet achieved consistency or alignment with organization-wide standards.

Defined: The architectural element is undergoing structured development within a comprehensive and standardized architectural

framework. Practices are now formally adopted, and the element has a clear alignment with strategic business imperatives. A dedicated architecture team is actively involved in its evolution.

Managed: The architecture element has reached a level of maturity where key performance indicators are monitored and operational effectiveness is actively reviewed. Collaboration between the architecture team and executive stakeholders is focused on identifying opportunities for refinement and improving return on investment.

Measured: In this final phase, the architecture element is considered a mature and optimized asset within the architecture portfolio. It serves as a reference model for similar initiatives. Emphasis is placed on leveraging data analytics to inform continuous improvement cycles and guard against performance plateau.

A fully completed maturity model based on the Architecture Capability Maturity Model framework is shown in Table 15-3.

Table 15-3. *Architecture Capability Maturity Model Framework*

	Initial	Under Development	Defined	Managed	Measured
Architecture process	Processes are ad hoc and localized. Some architecture processes are defined. Success depends on individual efforts.	The basic architecture process is clearly documented. The architecture process has developed clear roles and responsibilities.	The architecture is well-defined and communicated with operating unit IT responsibilities. The process is largely followed.	Architecture process is part of the culture. Quality metrics associated with the architecture process are captured.	Concerted efforts to optimize and continuously improve architecture process.
Architecture development	Architecture processes, documentation, and standards are established by a variety of ad hoc means and are localized or informal.	IT vision, principles, business linkages, Baseline Architecture, and Target Architecture are identified. Standards exist, but not necessarily linked to Target Architecture.	Gap analysis and migration plan are completed. Fully developed TRM and Standards Profile. IT goals and methods are identified.	Architecture documentation is updated on a regular basis to reflect the updated architecture. Architectures are defined by appropriate standards.	A standards and waivers process is used to improve architecture development process.

Business linkage	Minimal or implicit linkage to business strategies or business drivers.	Explicit linkage to business strategies.	Architecture is integrated with capital planning and investment control.	Capital planning and investment control are adjusted based on the feedback received and lessons learned. Periodic re-examination of business drivers.	Architecture process metrics are used to optimize and drive business linkages. Business involved in the continuous architecture process improvements.
Senior management involvement	Limited management team awareness or involvement in the architecture process.	Management awareness of architecture effort.	Senior management team aware of and supportive of the organization-wide architecture process. Management actively supports architectural standards.	Senior management team directly involved in the architecture review process.	Senior management involvement in optimizing process improvements in architecture development and governance.

(*continued*)

Table 15-3. (*continued*)

	Initial	Under Development	Defined	Managed	Measured
Business unit participation	Limited operating unit acceptance of the architecture process.	Responsibilities are assigned and work is underway.	Most elements of operating unit show acceptance of or are actively participating in the architecture process.	The entire operating unit accepts and actively participates in the architecture process.	Feedback on architecture process from all operating unit elements is used to drive architecture process improvements.
Architecture communication	The latest version of the operating unit's architecture documentation is online. Little communication exists about the architecture process and its improvements.	The architecture web pages are updated periodically and are used to document architecture deliverables.	Architecture documents updated regularly on the architecture web page.	Architecture documents are updated regularly and frequently reviewed for the latest architecture developments and standards.	Architecture documents are used by every decision-maker in the organization for every IT-related business decision.

Architecture governance	No explicit governance of architectural standards.	Governance of a few architectural standards and some adherence to existing Standards Profile.	Explicit documented governance of majority of IT investments.	Explicit governance of all IT investments. Formal processes for managing variances feed back into the architecture.	Explicit governance of all IT investments. A standards and waivers process is used to make governance process improvements.
IT security	IT security considerations are ad hoc and localized.	IT Security Architecture has defined clear roles and responsibilities.	IT Security Architecture Standards Profile is fully developed and is integrated with architecture.	Performance metrics associated with IT Security Architecture are captured.	Feedback from IT Security Architecture metrics is used to drive architecture process improvements.

(*continued*)

Table 15-3. (*continued*)

	Initial	Under Development	Defined	Managed	Measured
IT investment and acquisition strategy	Little or no involvement of strategic planning and acquisition personnel in the architecture process. Little or no adherence to existing standards.	Little or no formal governance of IT investment and acquisition strategy. Operating unit demonstrates some adherence to existing Standards Profile.	IT acquisition strategy exists and includes compliance measures to IT Architecture. Cost benefits are considered in identifying projects.	All planned IT acquisitions and purchases are guided and governed by the architecture.	No unplanned IT investment or acquisition activity.

Using the Architecture Capability Maturity Model framework provides a picture of how the IT Architecture fits with the rest of the organization. It clarifies where the organization stands in terms of applying the architecture to day-to-day operations and processes.

The Architecture Capability Maturity Model framework focuses primarily on IT architecture, which makes it less applicable to an Enterprise Architecture environment.

Something else that stands out about the Architecture Capability Maturity Model framework is the element called IT security. Judging by its name and content, it is focused solely on security in relation to IT. This contradicts what the TOGAF Standard said earlier about security being a cross-cutting concern (see Chapter 11, Section 11.2).

All in all, the Capability Maturity Model Integration framework described in the next section is a much better fit for determining and measuring Enterprise Architecture maturity.

15.1.2. Capability Maturity Model Integration

The use of a Capability Maturity Model Integration framework is preferred when it comes to defining and measuring Enterprise Architecture maturity because of its more strategic approach to viewing Enterprise Architecture. As mentioned in the previous section, the Architecture Capability Maturity Model framework puts a lot of emphasis on the IT side. Enterprise Architecture is a strategic discipline, and it is very different from IT Architecture. For this reason, it is more appropriate to use a Capability Maturity Model Integration framework to measure and provide insight into Enterprise Architecture maturity.

A maturity model based on Capability Maturity Model Integration (CMMI) defines five *elements* of Enterprise Architecture maturity. These are shown in Table 15-4.

Table 15-4. *Elements of CMMI Models*

Element	Description
Strategy and vision	A clear strategy and vision for architecture development that is closely aligned with the organization's goals is required.
Architecture governance	Well-trained, experienced, and competent people are responsible for developing the architecture.
Architecture method and process	The organization has standardized processes and methods for developing the architecture.
Architecture deliverables	The appropriate tools and technologies are available to design, implement, and manage the architecture.
Business alignment	There are clear controls and governance for architecture development, including measurable performance indicators and quality standards.

Strategy and Vision: The organization has a clear strategy and vision for architecture development that is closely aligned with the organization's goals. There are clear priorities and guidelines for architecture development.

Stakeholder involvement in the architecture is critical to the success of an Enterprise Architecture implementation. Stakeholders include not only architects, but also administrative roles such as senior (business and IT) management. Project and portfolio management play an important role in the implementation of an Enterprise Architecture as well. Ideally, there is support at all levels of the organization for the use of the architecture and the adoption of its results.

The Enterprise Architecture is evaluated on a regular basis and suggestions for improvement or further development are identified. In this way, the effectiveness of the Enterprise Architecture is continuously improved.

Architecture Governance: The organization has well-trained, experienced, and competent people responsible for developing the architecture. There are clear roles and responsibilities for the architecture function. These are clearly defined so that the entire organization understands what can and cannot be expected of the Architecture Capability. This prevents discussions and disagreements about the architecture.

Senior management training is organized so that the management level of the organization has sufficient substantive knowledge of Enterprise Architecture. The architecture discipline is also introduced during the onboarding process of new employees. Awareness programs are used to make employees aware of Enterprise Architecture.

Opportunities for collaboration with other similar organizations are actively sought. Experiences and ideas are exchanged to improve and develop the Enterprise Architecture.

Architecture Method and Process: The organization has standardized processes and methods for developing the architecture.

The architecture process, like any other process, must be maintained. This ensures the effectiveness and efficiency of the Enterprise Architecture. Captured metrics can be used proactively to identify and make improvements to the architecture process, framework, or products.

However, developing and implementing an Enterprise Architecture is not an end in itself. Going back to the earlier definition of Enterprise Architecture (see Chapter 4), the discipline ensures that an organization can implement the necessary initiatives to achieve its intended goals and objectives. Not surprisingly, this is the purpose of Enterprise Architecture.

In practice, of course, the use and application of architecture can vary. For example, it can be used simply as an information conduit (architecture in support of solution delivery) or as a means to drive individual projects (architecture in support of projects).

The ultimate goal of Enterprise Architecture is to be a tool that guides the entire organization in achieving business goals (architecture in support of strategy). The uses of the Architecture Capability are described in more detail in Chapter 9.

Architecture Deliverables: The organization has the appropriate tools and technologies to develop, implement, and manage the architecture.

Architecture is not just about creating a set of architecture deliverables (such as Architecture Principles, requirements, diagrams, and other models). Of course, architecture artifacts are very important to have, but they also need to be used and maintained. Maintaining architecture artifacts means updating them and, if necessary, removing artifacts that are no longer applicable. Active maintenance of architecture artifacts ensures that the architecture is and remains current and functional.

Working with architecture can be supported by the use of architecture tools. These tools must be well suited to the task for which they are used. The integrated use of tools, preferably with repository support, maximizes efficiency and effectiveness.

Business Alignment: The organization has clear controls and governance for architecture development, including measurable performance indicators and quality standards.

The ideal implementation of Enterprise Architecture is characterized by its use in support of strategy. Simply stating that projects must conform to the architecture is generally not enough. Ultimately, it is up to the organization to decide how to use an Enterprise Architecture (and that

may mean using it to manage the portfolio, some individual projects, or even occasionally to achieve specific solutions).

The evolution of the Enterprise Architecture and its alignment with the organization requires regular coordination with various stakeholders. Stakeholders such as senior management, information and project managers, and subject matter experts should be involved. For the architecture process to run smoothly, recurring and regular consultation with stakeholders is essential.

It is necessary to create and increase awareness of the Architecture Capability. This introduces the organization to the use and application of architecture concepts and processes.

A maturity model based on Capability Maturity Model Integration defines not only its *elements,* but also five *levels* of Enterprise Architecture maturity against which the Architecture Capability is measured. These are shown in Table 15-5.

Table 15-5. *Maturity Levels of CMMI Models*

Level	Name	Description
1	Ad hoc	Frameworks and standards have not yet been established, and the architecture process is generally conducted in an informal manner.
2	Repeatable	The architecture process is formalized, and there is a vision for the architecture.
3	Defined	The architecture framework to be used has been determined, and there is a formally defined process for adhering to the architecture process.
4	Managed	Measurement data is actively used to improve and refine the architecture process.
5	Optimal	The architecture process is mature; the organization has drivers, goals, and objectives aligned with the application of Enterprise Architecture.

Ad Hoc: Frameworks and standards have not yet been established, and the architecture process is generally conducted in an informal manner. The organization is aware of the benefits that Enterprise Architecture can provide, but has not yet defined a process to track and monitor the evolution of the architecture process. The organization relies on the knowledge and skills of individual employees. A business strategy exists, but has not yet been translated into implementation.

Repeatable: The architecture process is formalized, and there is a vision for the architecture. The architecture process is repeatable, and templates and standards are being developed. The need for adherence to architectural frameworks is recognized by senior management, and efforts are made to formalize it through process capture. Metrics are also being captured to evaluate the process. The organization's strategy is beginning to be translated into concrete goals and objectives. The words *enterprise* and *architecture* are used with some regularity in conversations at the strategic level.

Defined: The architecture framework to be used has been determined, and there is a formally defined process for adhering to the architecture process. The architecture framework is now formalized, and standards are widely used within the organization. There is a roadmap for the evolution of the architecture, and related activities are performed in accordance with the roadmap. Metrics are maintained and monitored for the purpose of evolving the architecture process.

Managed: Measurement data is actively used to improve and refine the architecture process. Data is analyzed and used to drive architecture performance. The Enterprise Architecture is invariably used to inform the organization's strategy. Adherence to the architecture framework is a standard activity within the organization.

Optimal: The architecture process is mature; the organization has drivers, goals, and objectives aligned with the application of Enterprise Architecture. Continuous improvements are made to the architecture process, and projects and other initiatives can no longer proceed without Enterprise Architecture input. Work is also done outside the boundaries of one's own organization to improve and refine the architecture together with similar organizations. Experiences are shared and suggestions for improving processes are discussed.

A fully completed maturity model based on the Capability Maturity Model Integration framework is shown in Table 15-6.

Table 15-6. *Capability Maturity Model Integration Framework*

	Ad Hoc	Repeatable	Defined	Managed	Optimal
Strategy and vision	Architecture activities are not formally initiated and happen ad hoc.	The organization has started drafting a vision for EA.	EA deployment is clearly defined, including governance roles and responsibilities.	The EA is evaluated and adjustments are identified to improve the EA.	Action plans are proactively implemented to increase EA effectiveness based on measured data.
Architecture governance	The need to define processes and standards is recognized.	The need for governance has been identified.	Architecture governance is defined (a consultative body has been created).	The communication process is revised to improve EA activity.	The organization collaborates with similar organizations exchanging ideas to improve their EA.
Architecture method and process	Architecture processes are ad hoc and inconsistent.	The architecture method is beginning to be reused to capture crucial EA information.	Templates are used so that capture of information is consistent.	EA is used to guide organizational development.	Business influences technology and technology influences business.

Architecture deliverables	The documentation of business drivers, goals and objectives, architecture standards, etc. is not formally defined.	Drivers, goals, and objectives have been identified.	Classification of existing standards is consistent.	Documentation of drivers, goals, and objectives has become a standard activity.	Captured business and technology information is used to proactively identify technology that will improve business operations.
Business alignment	The organization recognizes that staff need to become more familiar with working with architecture.	The organization has started raising awareness and understanding of EA concepts and processes.	The organization starts to operate as a team, using the defined architecture and standards.	Staff throughout the organization have a good understanding of the Architecture Principles.	Departments work together as contributors to the architecture and its processes.

15.2. Maturing the Architecture Capability

After defining the desired framework for measuring the maturity of the Architecture Capability in the organization, the task is to further professionalize and mature the capability. The maturing of the capability takes place along the axis of the aforementioned elements of the framework.

This example is based on the Capability Maturity Model Integration framework. The framework has the following elements:

- Strategy and vision
- Architecture governance
- Architecture method and process
- Architecture deliverables
- Business alignment

The CMMI framework also suggests five maturity levels for each of the elements (see Table 15-5). Together with the identified current maturity level, it is possible to identify the actions or activities required for each element to move from a lower level to a higher level. Figure 15-1 shows an excerpt from Table 15-6 illustrating the *Strategy and vision* element for the Ad hoc, Repeatable, and Defined maturity levels.

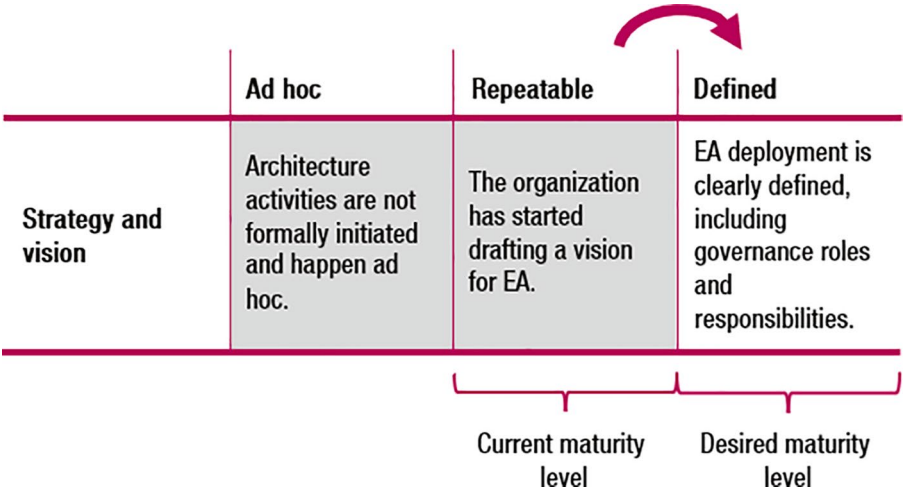


Figure 15-1. *Excerpt from the CMMI framework for the Strategy and vision element*

To move from the second level (Repeatable) to the next level (Defined), the vision for the use and application of the architecture must be supplemented with the necessary governance, roles, and responsibilities. This ensures that the architecture work has a sponsor from the executive level of the organization.

Questions that can be asked at this point are as follows:

- Has ownership of the architecture been assigned at the executive level?
- Can an owner be identified for the architecture processes?
- Is ownership of the architecture invested at the top management level?
- Is architecture also the responsibility of business management?
- Is the architecture owner held accountable for the contribution of the architecture to the business goals?

To answer these (and other) questions, it is important (among other things) to clarify the necessary responsibilities and accountability. One of the core components of doing architecture work properly is being able to establish ownership of the architecture and responsibility for its content. Table 15-7 lists the activities that can be invested in to achieve this.

Table 15-7. *List of activities to raise the maturity level of the Strategy and vision element*

Activity	Description
Obtain mandate for architecture	Ask senior management to express their commitment to architecture and explicitly assign responsibility for architecture.
Create responsibility matrix	Create a responsibility (RACI) matrix that maps architecture-related tasks to different functions within the organization.
Establish architecture board	Establish an architecture board to formally approve architecture products and provide an escalation platform.
Appoint architecture process owner	Delegate ownership of the architecture process. The process owner is responsible for the effectiveness and efficiency of the architecture processes.
Assign final responsibility for architecture	Ensure that senior management is effectively involved in architecture, especially on the business side.

After the above activities have been successfully implemented, the maturity level for the *Strategy and vision* element can be reviewed and reassessed. If the assessment is positive, the maturity level can be increased.

A second example of increasing maturity is to improve *the Architecture method and process* element. In Figure 15-2, this element is shown with the first three maturity levels.

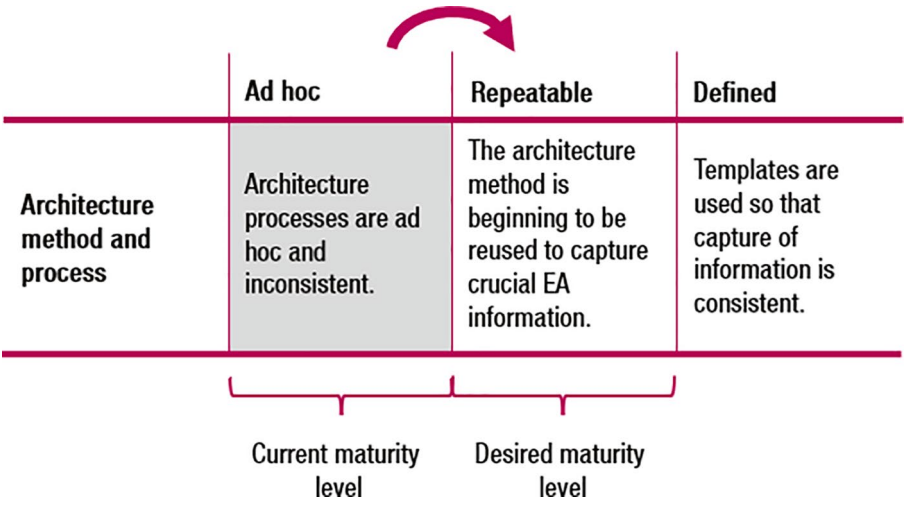


Figure 15-2. *Excerpt from the CMMI framework for the Architecture method and process element*

To move from the first level (Ad hoc) to the next level (Repeatable), it is necessary to examine how the architecture is created and applied in the organization. An indication that the maturity of the *Architecture method and process* element is at the *Ad hoc* level is that little or nothing has been done to formalize the architecture processes and their application.

Questions that can be asked at this point are as follows:

- Is an architecture only created when there is a client for it?
- Is an action plan created before an architecture is designed?
- Is the architecture process described?

- Is the architecture process known throughout the organization?
- Is there a regular review of whether the architecture process is still adequate?

After answering the above questions, the activities listed in Table 15-8 can be performed. Carrying out the listed activities will ensure that a higher level of maturity can also be achieved for this element.

Table 15-8. *List of activities to raise the maturity level of the Architecture method and process element*

Activity	Description
Plan architecture	Determine which architecture products are needed. Confirm this with stakeholders. Then create an action plan for creating the architecture.
Describe architecture processes	Describe the architecture processes, such as creating and maintaining the architecture, formalizing the architecture products, and the role of the architecture in projects.
Establish architecture product management process	Establish procedures for handling changes to the architecture. The goal is to actively maintain and update the architecture as a whole.

Once the activities listed in Table 15-8 have been performed and the processes and procedures are in place, it is possible to reassess whether the architecture is now at a higher level of maturity. If so, the maturity model can be updated with a higher rating.



Identify the Current Maturity Level: Begin by assessing where each element currently stands within the five maturity levels defined by the CMMI framework. This provides a clear starting point for improvement.

Determine the Next Desired Level: Decide on the next maturity level to achieve for each element. This sets a clear goal for the improvement efforts.

Ask the Right Questions: To understand what changes are needed, ask targeted questions about each element. This helps identify specific activities required to progress to the next level.

Implement the Necessary Activities: Based on the insights gained, carry out the identified activities. These actions are designed to address gaps and enhance the maturity of each element.

Review and Reassess: After implementing the activities, evaluate the changes to determine if the maturity level has improved. If the assessment is positive, acknowledge the advancement to the higher maturity level.

The listed actions and activities that can be undertaken to increase maturity for each element are just a few examples [25]. It is advisable to determine the actions in consultation with senior management and other key stakeholders.

15.3. Alignment with Competencies

The previous sections illustrated how a maturity framework and model can be used to determine the Architecture Capability maturity. To take this maturity to the next level, and thus properly implement the further professionalization of the capability in an organization, appropriate competencies are required.

Chapter 10, Section 10.4, describes the skills and competencies for each core architecture role. Table 15-9 shows how these competencies map to the elements of the Capability Maturity Model Integration framework.

Table 15-9. *CMMI elements mapped to competencies*

Element	Competency	Description
Strategy and vision	Strategic thinking and vision	Ability to conceptualize the long-term view of enterprise systems and their alignment with organizational goals.
	Strategic decision-making	Making informed, strategic decisions under uncertainty or complex scenarios.
Architecture governance	Leadership and influence	Architects need to have the capability to guide and inspire teams toward shared goals.
	Governance and compliance	Focuses on enforcing standards, policies, and guidelines that ensure architectural consistency.
Architecture method and process	Communication and collaboration	Ability to communicate complex ideas clearly to both technical and non-technical audiences.
	Adaptability and resilience	The ability to maintain composure and thus effectiveness in high-pressure situations.
	Analytical thinking and problem solving	Ability to break down complex challenges into manageable components.
Architecture deliverables	Systems thinking	Understanding the interconnectedness of components within an enterprise ecosystem is an absolute must.
Business alignment	Change leadership	Driving organization-wide change initiatives with resilience and adaptability.
	Business and technology alignment	Architects must be able to bridge the gap between business strategy and technology implementation.

Table 15-9 shows, between the lines, which architectural roles are required to fulfill each of the elements in the maturity model. These roles, along with the elements and competencies, are shown in Table 15-10.

Table 15-10. *CMMI elements, competencies, and core architecture roles*

Element	Competency	Core Architecture Role
Strategy and vision	Strategic thinking and vision	Enterprise Architect, Segment Architect
	Strategic decision-making	Enterprise Architect, Segment Architect
Architecture governance	Leadership and influence	Enterprise Architect
	Governance and compliance	Enterprise Architect, Segment Architect, Solution Architect
Architecture method and process	Communication and collaboration	Enterprise Architect, Segment Architect, Solution Architect
	Adaptability and resilience	Enterprise Architect, Segment Architect, Solution Architect
	Analytical thinking and problem solving	Enterprise Architect, Segment Architect, Solution Architect
Architecture deliverables	Systems thinking	Segment Architect, Solution Architect
Business alignment	Change leadership	Enterprise Architect, Segment Architect
	Business and technology alignment	Enterprise Architect, Segment Architect

The tables showing the mapping between the elements of the CMMI framework on the one hand and the required competencies and core architecture roles on the other show that several roles are needed to take the Architecture Capability within an organization to the next level. It is clear that having only one architecture role in an organization is not enough to significantly increase maturity.

It is also clear from the mapping that the role of the Enterprise Architect has a significant stake in further professionalizing the Architecture Capability.

The TOGAF Standard does not (yet) have its own maturity model. It is not inconceivable that one will be included in a later edition of the framework. Hopefully, a maturity model will be chosen that focuses on the more strategic nature of the discipline and less on the IT domain.

A maturity model is a powerful tool. Its use enables an organization to identify opportunities for growth and development. By translating the activities associated with the maturity elements into actionable initiatives, it becomes possible to increase the level of architecture maturity within the organization.



Further Reading

The TOGAF Standard Series Guide "Architecture Maturity Models", document number G203, offers additional information about the use of maturity models.

An interesting article on architecture maturity models and how to read them can be found at the following link:

https://www.ardoq.com/knowledge-hub/enterprise-architecture-maturity-model

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15.4. Summary

Maturity models are commonly used to measure and provide insight into the maturity of the internal Architecture Capability. The TOGAF Standard highlights two well-known maturity frameworks.

The *Architecture Capability Maturity Model* framework has a structure in which the *core components of an architecture process* are represented and measured, often in relation to the organization's IT. The purpose of using such a model is to identify weaknesses and provide a step-by-step approach to improving them.

The Architecture Capability Maturity Model considers the following nine architecture elements when assessing architecture maturity:

- Architecture process
- Architecture development
- Business linkage
- Senior management involvement
- Operating unit participation
- Architecture communication
- Architecture governance
- IT security
- IT investment and acquisition strategy

The *Capability Maturity Model Integration* framework, on the other hand, places more emphasis on maturing the Architecture Capability in order to *achieve organizational goals*. This makes this type of maturity model more appropriate for use in an Enterprise Architecture environment.

A maturity model based on Capability Maturity Model Integration defines five elements of Enterprise Architecture maturity:

- Strategy and vision
- Architecture governance
- Architecture method and process
- Architecture deliverables
- Business alignment

Appropriate competencies are needed to properly implement the further professionalization of the Architecture Capability in an organization.

In Chapter 10, Section 10.4, architecture roles were mapped to competencies. This chapter maps these competencies (and their corresponding architecture roles) to the elements of the Capability Maturity Model Integration framework.

In order to take the Architecture Capability within an organization to the next level, several architecture roles are needed in an organization. The role of the Enterprise Architect has a significant stake in further professionalizing the Architecture Capability.

A maturity model is a powerful tool. Its use enables an organization to identify opportunities for growth and development. By translating the activities associated with the maturity elements into actionable initiatives, it becomes possible to increase the level of architecture maturity within the organization.